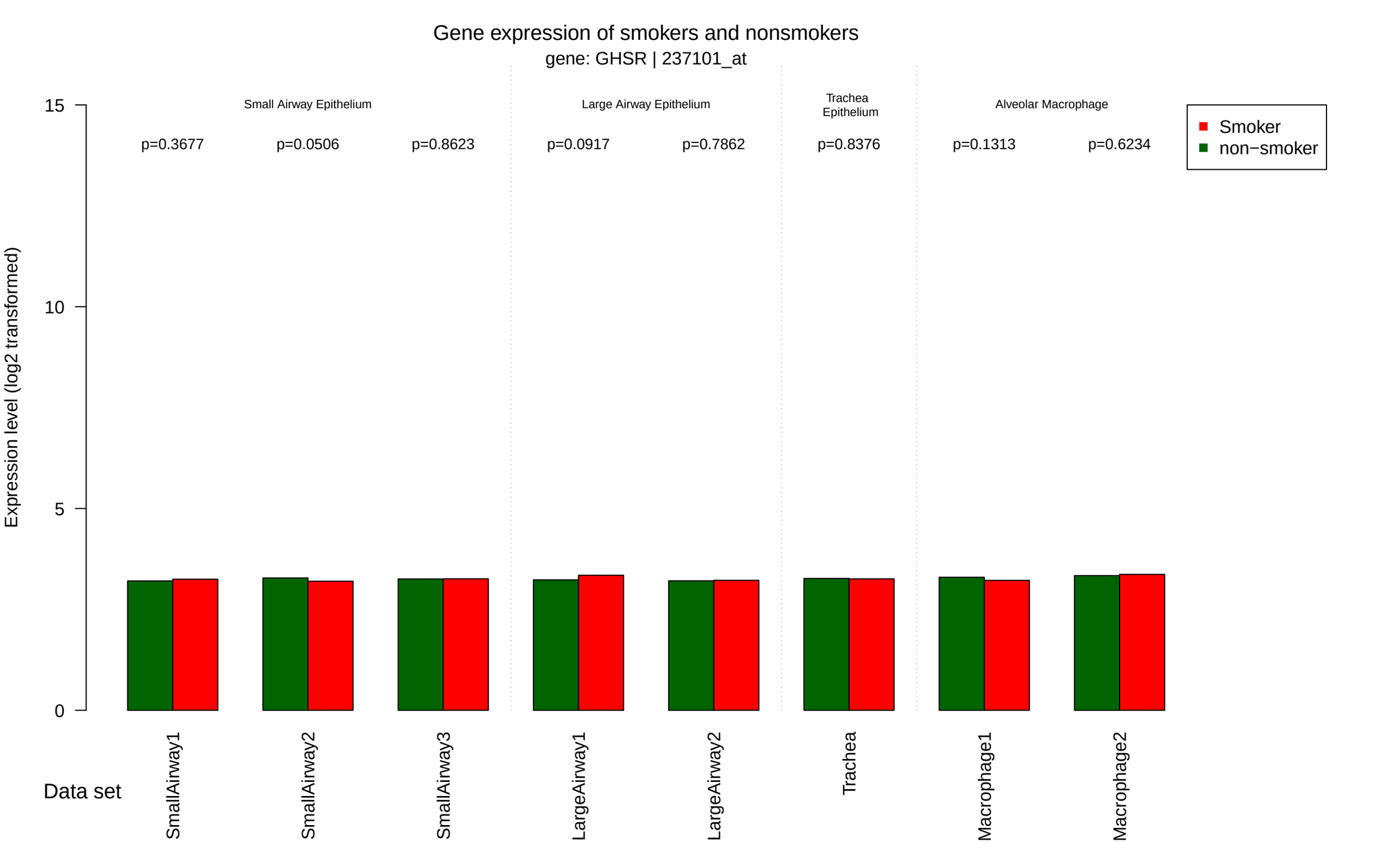
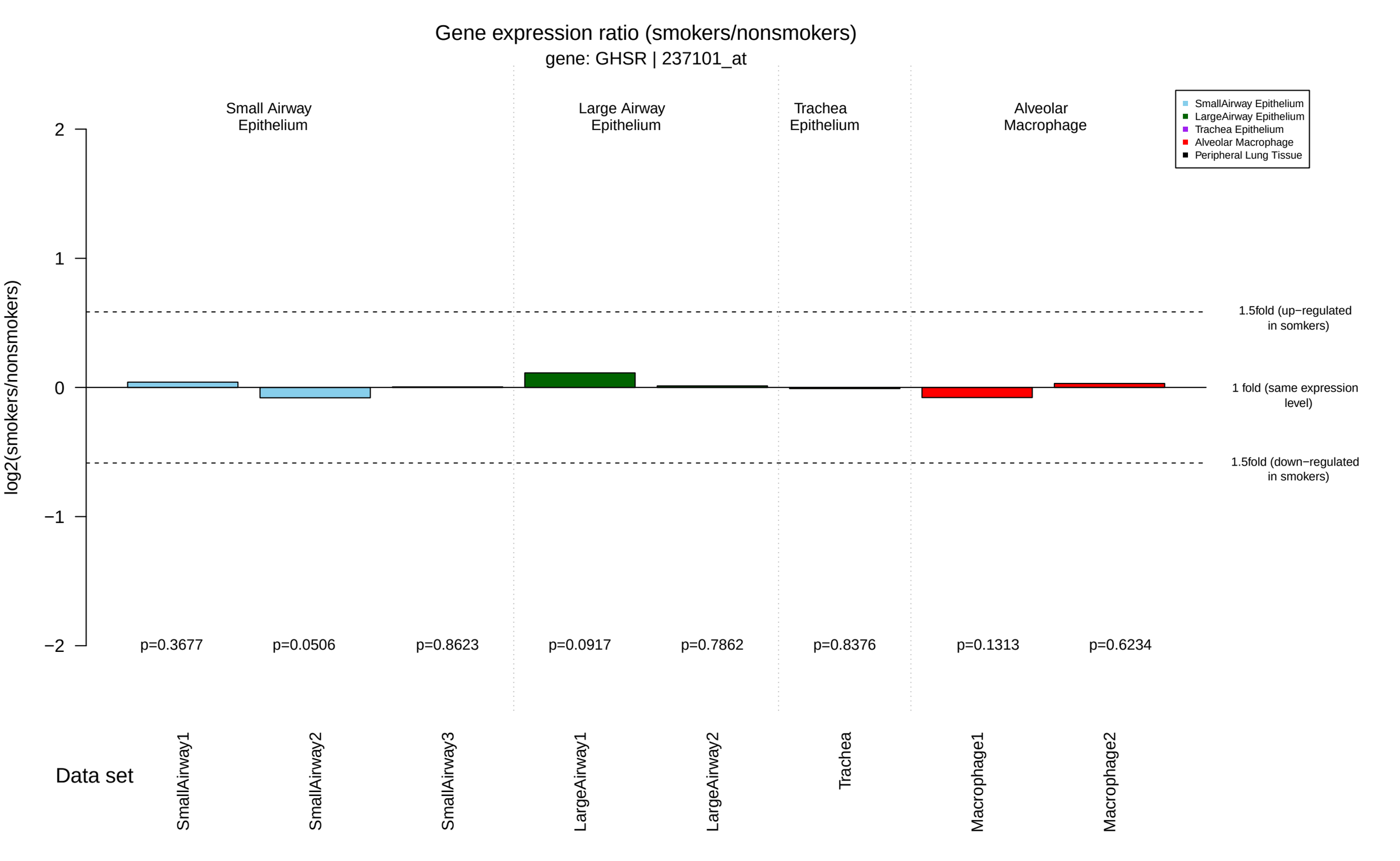
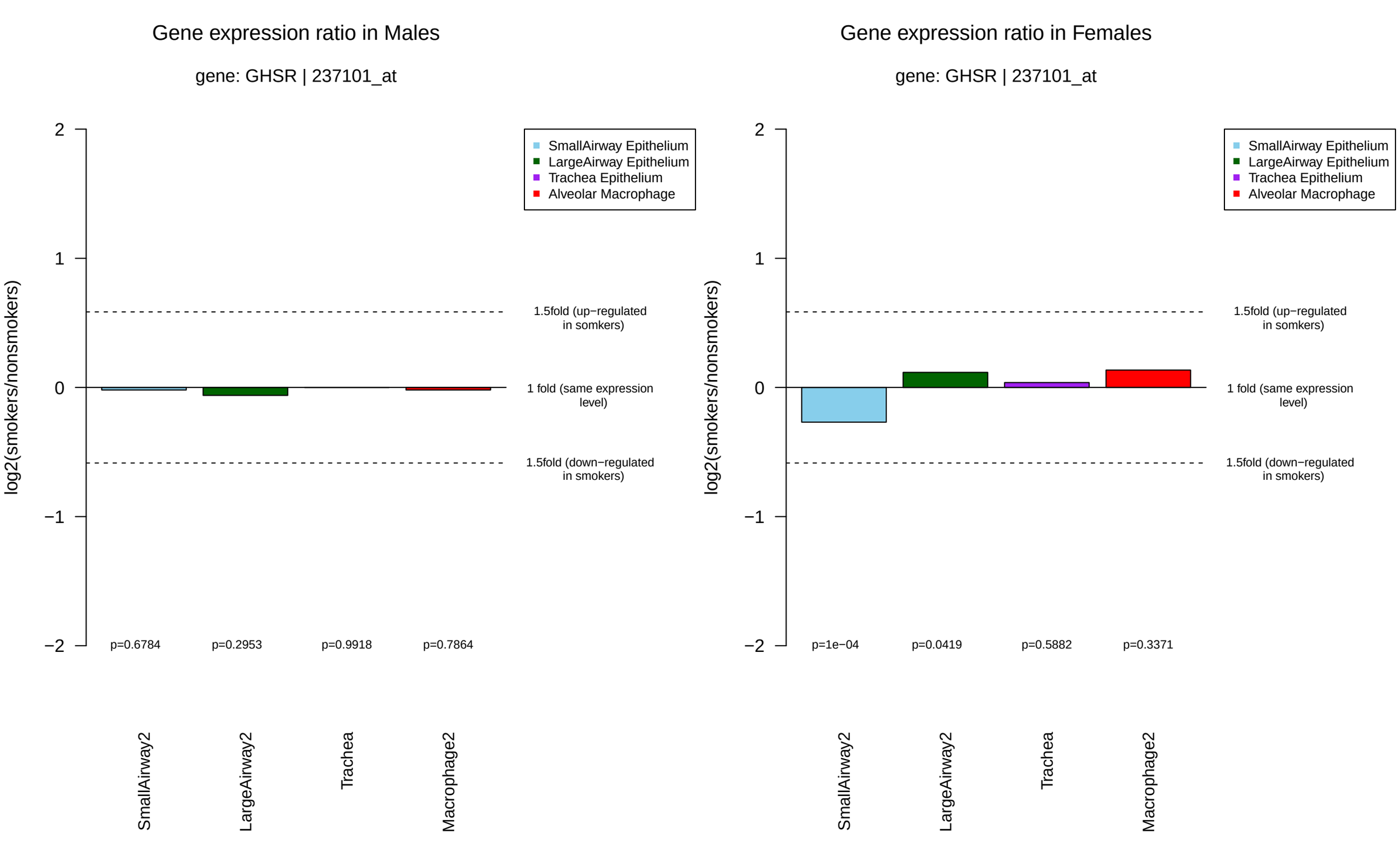
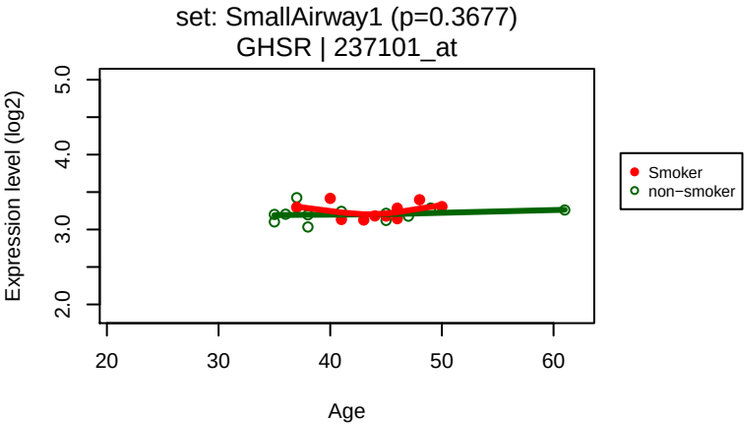
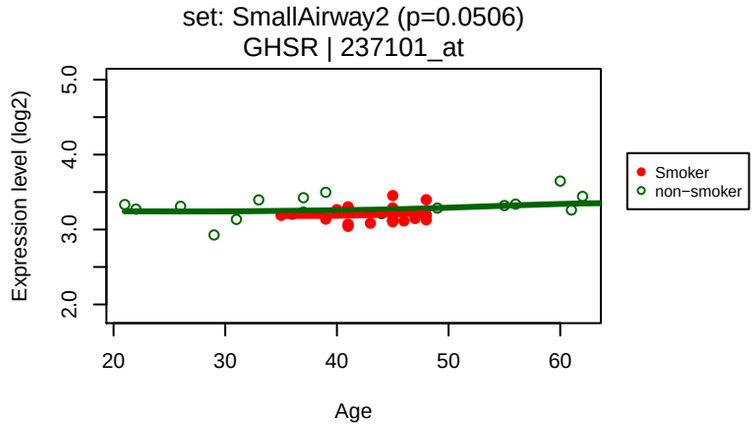
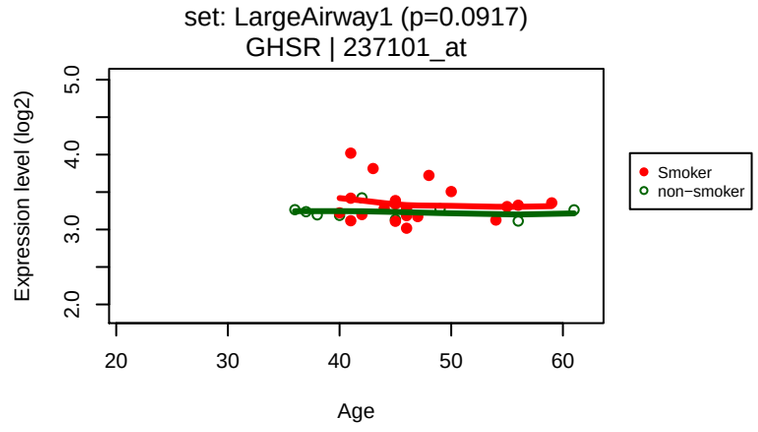

Gene expression of smokers and nonsmokers
gene: GHSR | 237101_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3677
p=0.0506
p=0.8623
p=0.0917
p=0.7862
p=0.8376
p=0.1313
p=0.6234
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

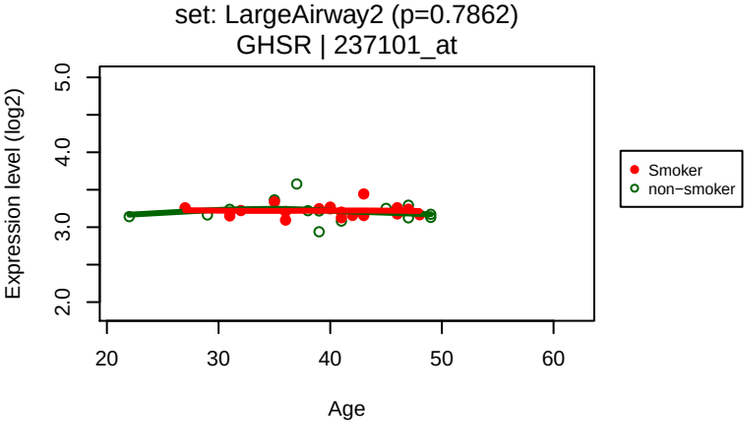

Gene expression ratio (smokers/nonsmokers)
gene: GHSR | 237101_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3677
p=0.0506
p=0.8623
p=0.0917
p=0.7862
p=0.8376
p=0.1313
p=0.6234
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

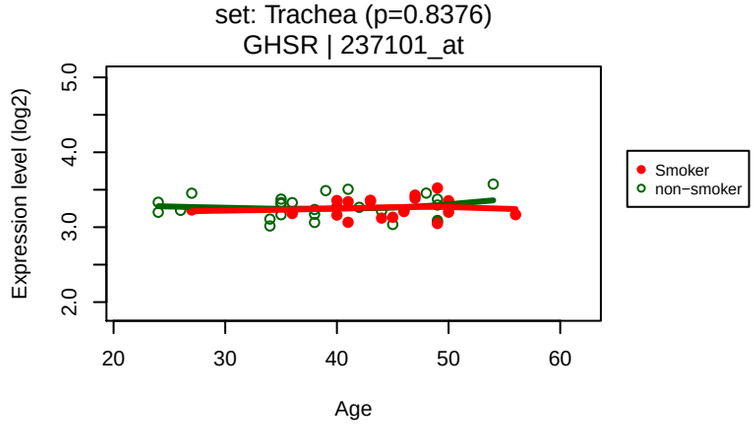

Gene expression ratio in Males
gene: GHSR | 237101_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6784
p=0.2953
p=0.9918
p=0.7864
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GHSR | 237101_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=1e−04
p=0.0419
p=0.5882
p=0.3371
SmallAirway2
LargeAirway2
Trachea
Macrophage2

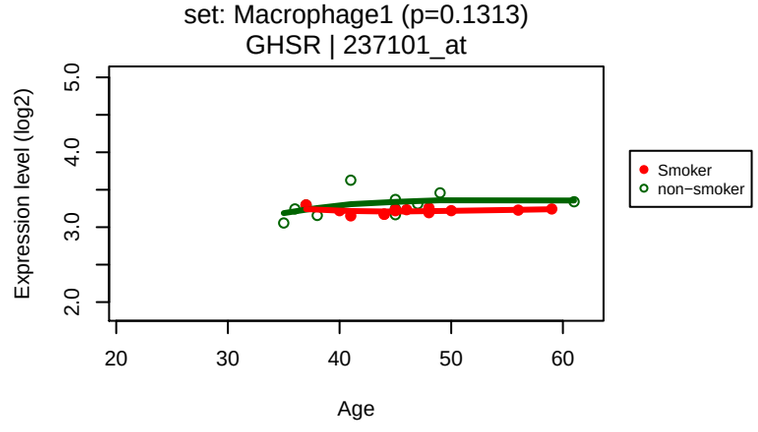

set: SmallAirway1 (p=0.3677)
GHSR | 237101_at
Expression level (log2)
Age
Smoker
non−smoker

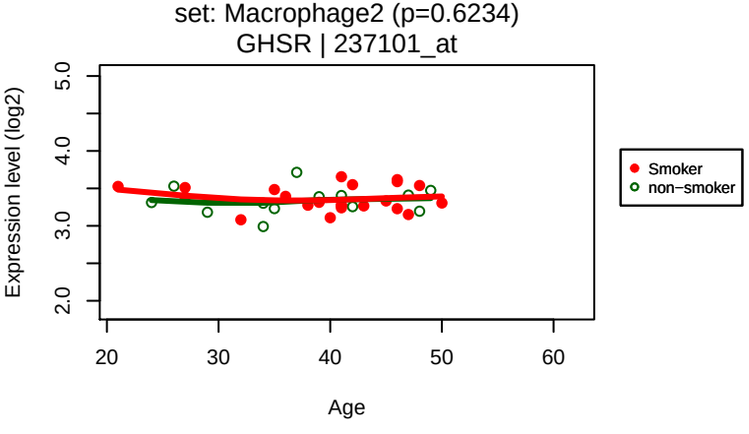

set: SmallAirway2 (p=0.0506)
GHSR | 237101_at
Expression level (log2)
Age
Smoker
non−smoker

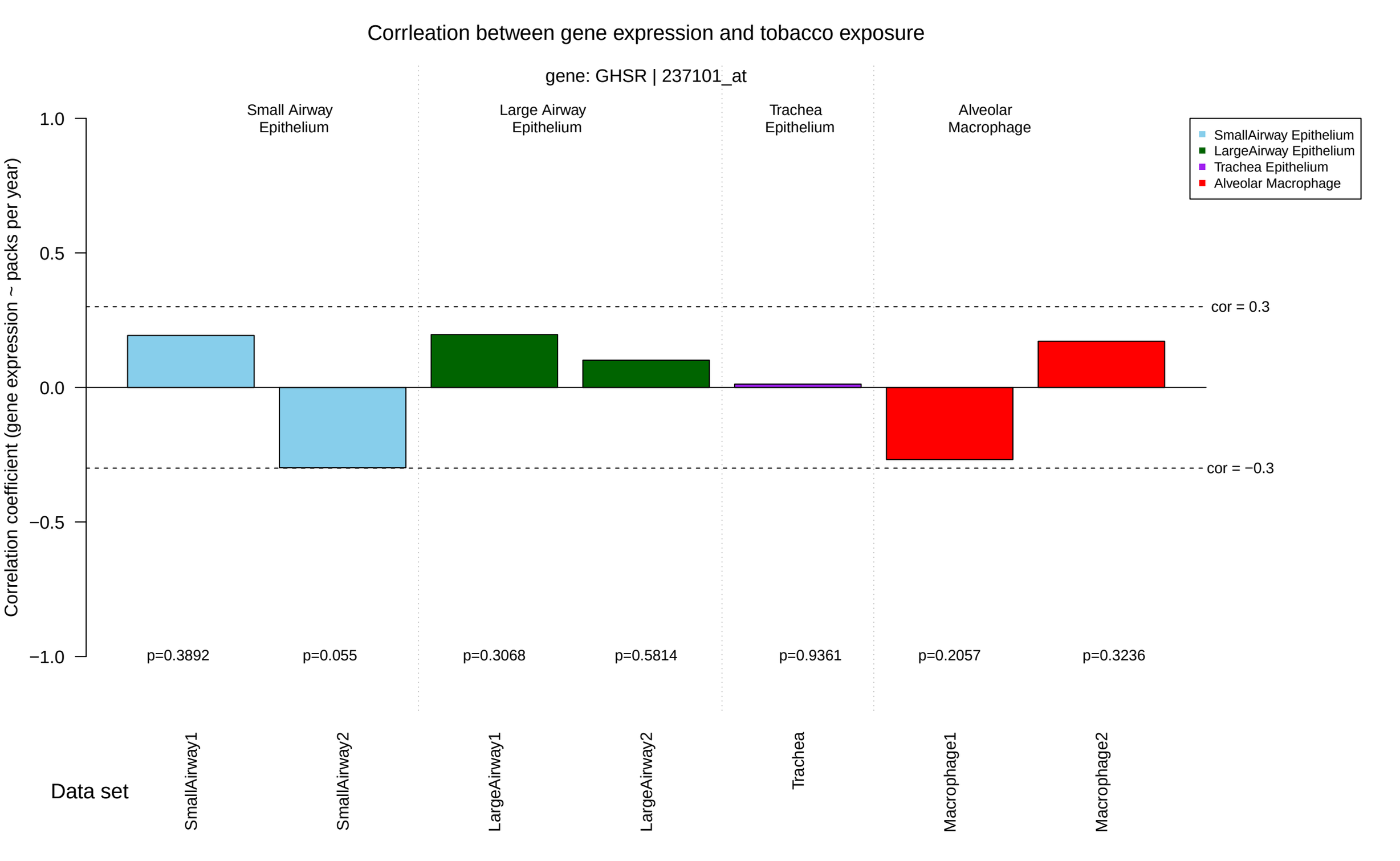

set: LargeAirway1 (p=0.0917)
GHSR | 237101_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7862)
GHSR | 237101_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8376)
GHSR | 237101_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1313)
GHSR | 237101_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6234)
GHSR | 237101_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GHSR | 237101_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3892
p=0.055
p=0.3068
p=0.5814
p=0.9361
p=0.2057
p=0.3236
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2